

Learning Preference Models from Choice Data

M.tech thesis Project
Bradley-Terry, Full CDM, and Low-rank CDM

Advisor: Prof. Suryanarayana Sankagiri

Nitesh Kumar Shah

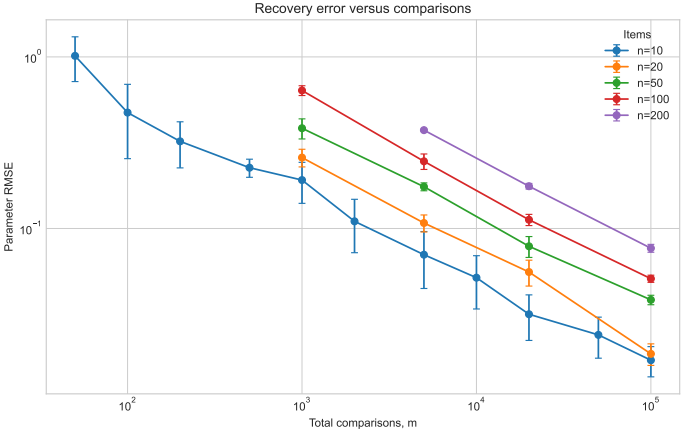
M.Tech I2MP Data Science and Artificial Intelligence, IIT Madras

Progress recap

Stage	Objective	Completed work
BTL	Recover pairwise strengths	Simulation, Choix estimation, recovery scaling, sampling, MLE diagnostics, and optimizer comparisons
Full CDM	Learn context effects	Full-matrix simulation and estimation. Observation scaling used $n = 5$ and 1,000–50,000 choices; item scaling used $n = 5$–80 with 50,000 choices
Cross fitting	Measure model mismatch and examine misspecification	BTL data fitted with CDM and CDM data fitted with BTL under controlled splits
Low-rank CDM	Control model complexity	Ranks 1–3 with three restarts per rank on the same data but different initialization seeds; rank selection by validation NLL and held-out NLL evaluation

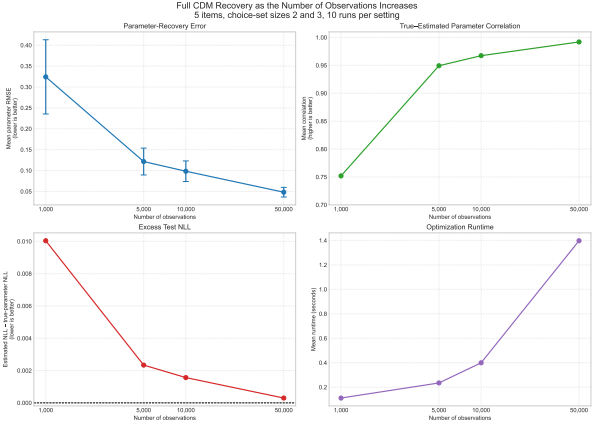
I am updating code, experiment figures, and the thesis skeleton in Overleaf.

Bradley–Terry recovery improves with data



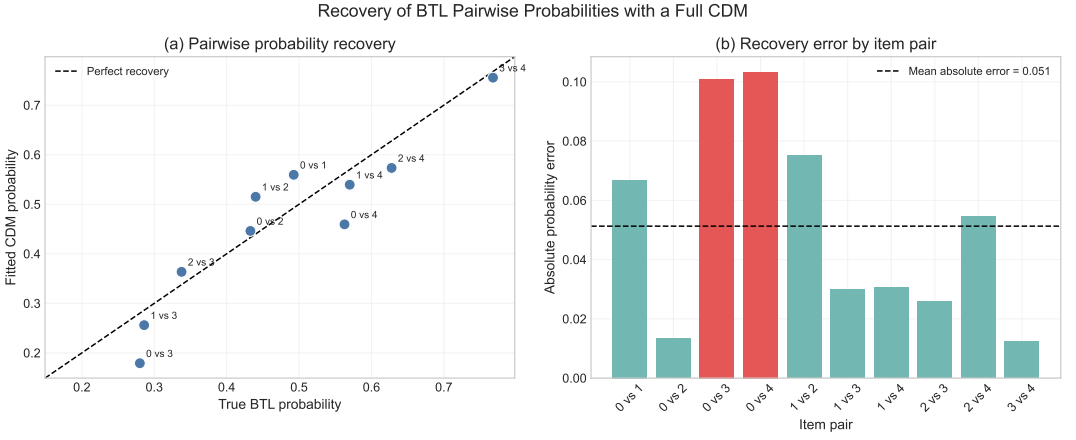
Note: More comparisons reduce recovery error and trial-to-trial variation.

Full CDM recovery improves with observations



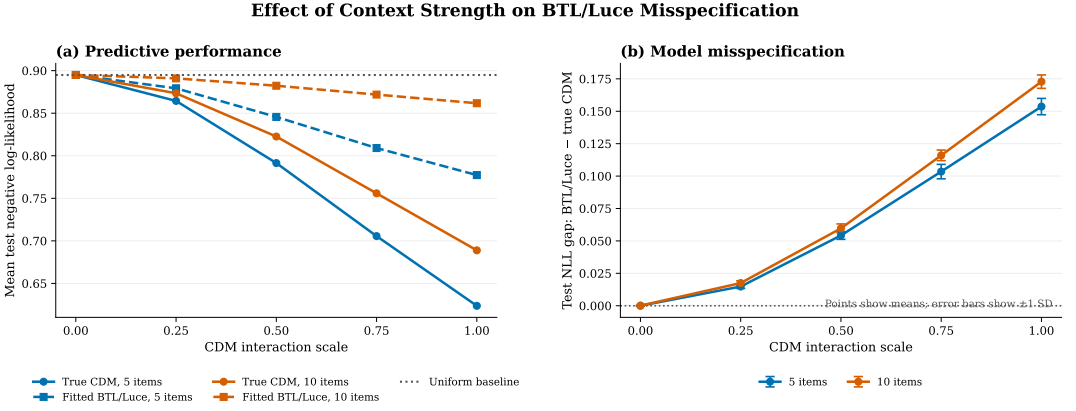
Note: Recovery improves as the number of observed choices grows.

BTL-generated data fitted with a full CDM



Result: Full CDM matches BTL pairwise probabilities, but pairwise data do not identify unique CDM parameters.

CDM-generated data fitted with BTL/Luce



Result: Ignoring context produces a positive held-out NLL gap that grows with interaction strength.

BTL/Luce misspecification

Held-out misspecification gap

$$\Delta_{\mathrm{NLL}} = L_{\mathrm{test}}(\hat{\theta}_{\mathrm{BTL}}) - L_{\mathrm{test}}(\theta_{\mathrm{CDM}}).$$

$\Delta_{\mathrm{NLL}} \approx 0$ means BTL is sufficient. $\Delta_{\mathrm{NLL}} > 0$ measures the predictive cost of ignoring context.

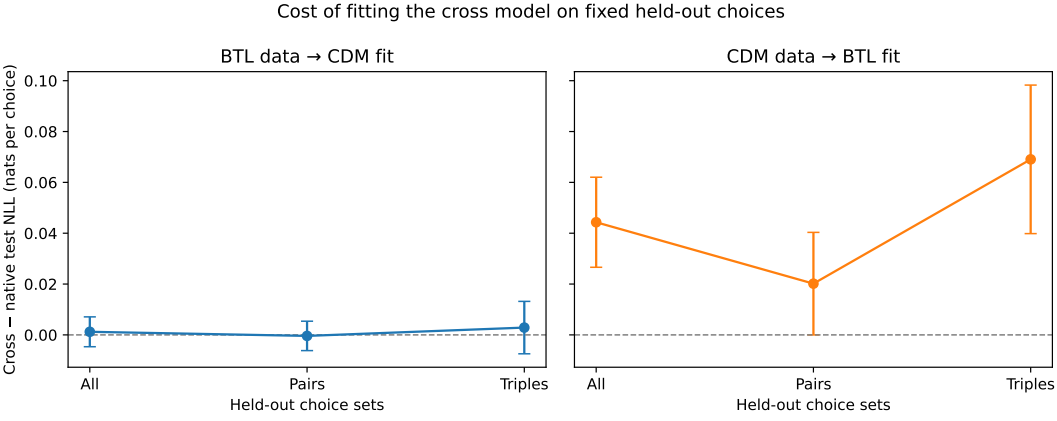
Mean test NLL across 10 runs

Items	Interaction scale	True CDM	Fitted BTL	Δ_{NLL}
5	0.00	0.8949	0.8949	0.0000
5	0.50	0.7914	0.8456	0.0542
5	1.00	0.6237	0.7773	0.1536
10	0.00	0.8949	0.8949	0.0001
10	0.50	0.8226	0.8823	0.0597
10	1.00	0.6889	0.8617	0.1728

Controlled setting: 50,000 observations, choice-set sizes 2 and 3, common train/test protocol.

Stronger context effects produce a larger predictive penalty for BTL/Luce.

Held-out loss separates native and mismatched fits



Interpretation: Test NLL gaps quantify predictive loss under model misspecification.

Low-rank CDM reparameterization

Full CDM

$$u_i(S) = \sum_{\substack{j \in S \\ j \neq i}} A_{ij}$$

$$P(i \mid S) = \frac{e^{u_i(S)}}{\sum_{k \in S} e^{u_k(S)}}$$

$A \in \mathbb{R}^{n \times n}$ stores a directed context effect for each ordered item pair.

Parameter scale: $O(n^2)$

Low-rank CDM

$$A = TC^{\mathsf{T}}$$

$$A_{ij} = t_i^{\mathsf{T}} c_j = \sum_{k=1}^{r} t_{ik} c_{jk}$$

Item i has target vector $t_i \in \mathbb{R}^r$ and context vector $c_i \in \mathbb{R}^r$.

Parameter scale: $O(nr)$

Rank r controls the number of latent interaction directions represented by the model.

Rank selection and uniform performance guarantee

Rank-selection procedure

1. Fit each candidate rank using multiple initialization seeds.
2. Retain converged solutions.
3. Select rank using validation NLL.
4. Evaluate the selected model once on test data.

Validation controls capacity without reusing the held-out test set.

Uniform performance guarantee

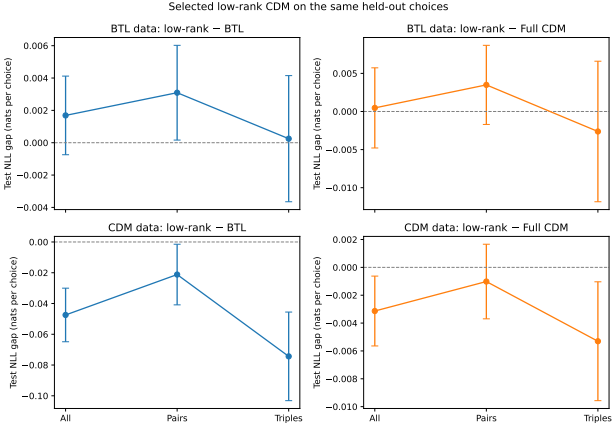
$$\Pr\left(\sup_{\theta\in\Theta_r}\left|\widehat{L}_m(\theta)-L(\theta)\right|\le\varepsilon\right)\ge 1-\delta.$$

With probability at least $1-\delta$, empirical loss stays within ε of population loss for every model in the rank-r class Θ_r.

More observations tighten the bound. Lower rank reduces model complexity.

Exact constants depend on the assumptions in Seshadri et al.

Current low-rank experiment



Ranks 1–3, three restarts per rank on the same data with different initialization seeds.

Next steps

Complete the low-rank study

1. Extend the grid to ranks 4 and 5.
2. Diagnose non-converged runs.
3. Select rank using validation NLL.
4. Evaluate once on held-out test data.

Complete the comparison

1. Compare full and low-rank CDM run-time.
2. Compare test NLL and parameter count.
3. Reproduce the paper's testing experiments.
4. Also to write and add verified figures and tables to Overleaf.

Immediate priority: establish a reliable rank-versus-performance curve and understand Uniform performance guarantee in depth